

BREAST CANCER DETECTION USING SUPPORT VECTOR MACHINE ON A CUSTOM SYNTHETIC REGIONAL DATASET OF JHARKHAND

(Machine Learning AI26509)

An Assignment Report Submitted to
Dr. Debjani Mustafi

SUBMITTED BY

ROHIT KUMAR MANDAL
(Roll No: MT/CS/10013/26)

RAJ PRAKASH KUMAR MAHTO
(Roll No: MT/CS/10002/26)

BIRLA INSTITUTE OF TECHNOLOGY
MESRA-835215, RANCHI

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Abstract

Breast cancer classification is an important application of machine learning in healthcare research. Clinical records can be represented through numerical and categorical features that can be analyzed to distinguish between benign and malignant tumor categories.

This project presents a Support Vector Machine (SVM)-based classification approach using a custom synthetic dataset containing 569 records representing six regions of Jharkhand: Ranchi, Jamshedpur, Dhanbad, Dumka, Hazaribagh, and Bokaro. Five input features are used: Age, Tumor Size, Lymph Node Status, Family History, and Body Mass Index (BMI).

The dataset is divided into training and testing subsets using an 80:20 split. StandardScaler is applied to the input features, and an SVM classifier with a Radial Basis Function (RBF) kernel is trained using $C = 1.0$ and $\gamma = \text{scale}$. A Decision Tree classifier is used as a baseline model for comparison.

The SVM model achieves an accuracy of 88.60%, while the Decision Tree baseline achieves 84.21% accuracy. The SVM model also obtains 81.63% precision, 90.91% recall, and 86.02% F1-score for the Malignant class.

The experimental results demonstrate the performance of the SVM classifier on the selected synthetic dataset. However, the dataset does not contain real patient records, and the results should not be interpreted as clinical validation.

1 Introduction

1.1 Background and Motivation

Machine Learning has become an important area of research in healthcare data analysis. It provides computational methods for identifying patterns in structured datasets and performing classification tasks.

Breast cancer classification is one such application. A tumor can be categorized as benign or malignant based on relevant clinical features. Machine learning algorithms can be trained to identify patterns in these features and produce predictions for previously unseen records.

Support Vector Machine (SVM) is a supervised learning algorithm that identifies a decision boundary separating different classes. It attempts to maximize the margin between the classes and can handle non-linear classification problems through kernel functions.

In this project, SVM with a Radial Basis Function (RBF) kernel is used as the primary classification algorithm. A Decision Tree classifier is included as a baseline for performance comparison.

1.2 Literature Review and Research Gaps

Several studies have explored Support Vector Machine applications in classification and medical data analysis.

Zhou et al. (2022) [1] discussed SVM-based mining of electronic medical records for disease diagnosis. Their work provides background on the use of classification techniques in medical data.

Abdelfattah et al. (2023) [2] presented a privacy-preserving medical diagnosis system based on Support Vector Machine techniques.

Guido (2024) [3] provided an overview of advancements in Support Vector Machine methods, including the use of kernel functions for classification.

Huynh et al. (2024) [4] explored the application of SVM in rotating machinery fault detection using feature extraction techniques. This study provides an example of SVM-based classification in a different application domain.

Research Focus:

The present project focuses on implementing an RBF-based SVM classifier using a custom synthetic dataset representing six regions of Jharkhand. It also evaluates the classifier against a Decision Tree baseline using the same train-test split.

1.3 Why Machine Learning for Classification?

- **Pattern Recognition:** Machine learning models can identify relationships between input features and target classes.
- **Consistency:** A trained model applies the same learned classification procedure to different input records.
- **Automated Classification:** Machine learning can classify structured records without manually defining every decision rule.

- **Experimental Analysis:** Different algorithms can be evaluated using standard performance metrics.

1.4 Problem Statement

The objective of this project is to develop a machine learning-based system that classifies synthetic tumor records into Benign and Malignant categories.

The system is designed to:

1. Select relevant input features from the dataset.
2. Preprocess and scale the input features.
3. Train an SVM classifier using an RBF kernel.
4. Evaluate the classification performance using standard evaluation metrics.
5. Compare the SVM classifier with a Decision Tree baseline.

1.5 Organization of the Report

Chapter 1 introduces the project, its motivation, and problem statement.

Chapter 2 explains the technical background of SVM, kernel functions, and classification evaluation metrics.

Chapter 3 describes the project objectives and system architecture.

Chapter 4 presents the mathematical methodology and Python implementation.

Chapter 5 describes the dataset and experimental results.

Chapter 6 discusses the computational complexity of the proposed approach.

Chapter 7 presents the conclusion, limitations, and future scope.

2 Technical Background

2.1 Support Vector Machines (SVM)

Support Vector Machine is a supervised learning algorithm commonly used for classification tasks.

It attempts to find an optimal hyperplane that separates data points belonging to different classes.

For a two-class classification problem, the decision boundary can be represented as:

$$w^T x + b = 0 \tag{1}$$

where:

- w represents the weight vector.
- x represents the input feature vector.
- b represents the bias term.

The data points closest to the decision boundary are called support vectors. These points influence the position of the separating hyperplane.

2.2 Maximum Margin and Kernel Trick

SVM attempts to maximize the margin between the decision boundary and the nearest data points.

For non-linearly separable data, the kernel trick allows SVM to construct non-linear decision boundaries without explicitly calculating the transformed high-dimensional feature vectors.

2.3 Radial Basis Function Kernel

This project uses the Radial Basis Function (RBF) kernel.

It is mathematically represented as:

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right) \tag{2}$$

where:

- x_i and x_j represent input feature vectors.
- γ controls the influence of individual training samples.
- $K(x_i, x_j)$ represents the kernel similarity.

The RBF kernel is used to model non-linear relationships in the input feature space.

2.4 Feature Scaling

Feature scaling is important for SVM because input features may have different numerical ranges.

StandardScaler transforms each feature using:

$$z = \frac{x - \mu}{\sigma} \tag{3}$$

where:

- x is the original feature value.
- μ is the training-set mean.
- σ is the training-set standard deviation.
- z is the standardized value.

The scaler is fitted using the training data and then applied to the testing data.

2.5 Evaluation Metrics

The performance of the classification models is evaluated using the following metrics.

Accuracy:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

Precision:

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

Recall:

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

F1-Score:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{7}$$

where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

In this project, Malignant is treated as the positive class for calculating precision, recall, and F1-score.

3 Project Goals and System Architecture

3.1 Objectives

The major objectives of the project are:

1. Study the theoretical foundations of Support Vector Machine.
2. Understand the working of the RBF kernel.
3. Use a custom synthetic dataset representing six regions of Jharkhand.
4. Implement an SVM classifier using Python and Scikit-learn.
5. Use Decision Tree as a baseline classifier.
6. Evaluate both models using accuracy, precision, recall, and F1-score.
7. Generate graphical representations of the dataset and model performance.

3.2 High-Level System Architecture

The proposed system follows a machine learning classification pipeline consisting of the following stages.

1. **Dataset Loading:** Load the synthetic Jharkhand dataset from a CSV file.
2. **Data Preprocessing:** Select five input features and encode the target variable into numerical values.
3. **Train-Test Splitting:** Divide the dataset into 80% training data and 20% testing data.
4. **Feature Scaling:** Apply StandardScaler to the SVM input features.
5. **Model Training:** Train SVM with an RBF kernel and Decision Tree as a baseline model.
6. **Performance Evaluation:** Calculate accuracy, precision, recall, F1-score, and confusion matrix.
7. **Visualization:** Generate five figures for dataset and classification analysis.

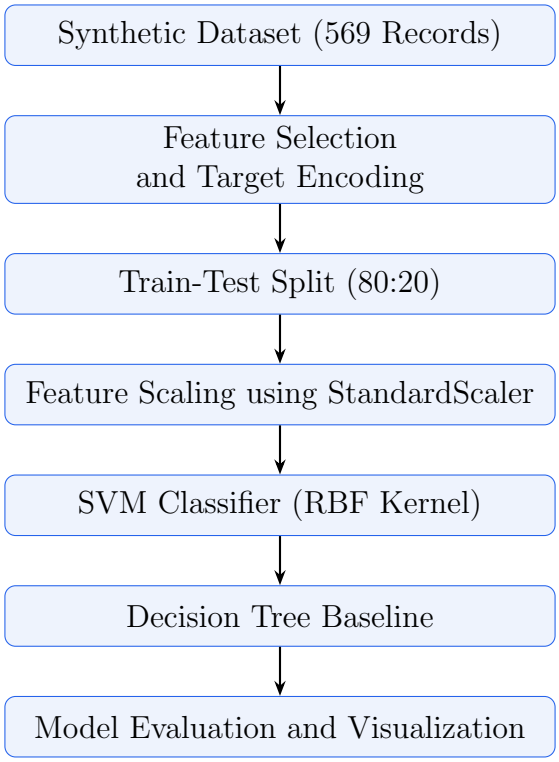


Figure 1: Overall architecture of the proposed classification system.

4 Methodology and Implementation

4.1 Mathematical Formulation of SVM

The objective of SVM is to identify a separating hyperplane that maximizes the margin between different classes.

For a soft-margin SVM, the optimization problem is expressed as:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i \tag{8}$$

Subject to:

$$y_i(w^T x_i + b) \geq 1 - \xi_i \tag{9}$$

$$\xi_i \geq 0 \tag{10}$$

where:

- w represents the weight vector.
- b represents the bias.
- C is the regularization parameter.
- ξ_i represents slack variables.
- n is the number of training samples.

The regularization parameter controls the trade-off between margin width and classification errors.

4.2 Custom Synthetic Dataset

The project uses a custom synthetic dataset containing 569 records representing six regions of Jharkhand.

The dataset is designed for experimental machine learning classification. It does not contain real patient records collected from hospitals.

The target variable is Diagnosis, which contains two categories: Benign and Malignant.

The dataset contains the following features:

- Age
- Tumor Size
- Lymph Node Status
- Family History
- Body Mass Index (BMI)

The classification target is encoded as:

$$Benign = 0 \tag{11}$$

$$Malignant = 1 \tag{12}$$

4.3 Core Algorithm Implementation in Python

The project was implemented using Python 3 and the Scikit-learn library.

Pandas is used for loading the CSV dataset. StandardScaler is used for feature scaling. The SVC classifier implements the RBF-based SVM, while DecisionTreeClassifier is used as the baseline.

The core implementation is presented in Listing 1.

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Load dataset
df = pd.read_csv("Jharkhand_Breast_Cancer_Dataset.csv")

# Feature selection and target encoding
X = df[['Age', 'Tumor_Size_mm', 'Lymph_Node_Status',
        'Family_History', 'BMI']]

y = df['Diagnosis'].map({
    'Benign': 0,
    'Malignant': 1
})

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Feature scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# Train SVM model
svm_model = SVC(
    kernel='rbf',
    C=1.0,
    gamma='scale'
)

```

```
svm_model.fit(X_train_scaled, y_train)
svm_pred = svm_model.predict(X_test_scaled)

# Train Decision Tree baseline
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
dt_pred = dt_model.predict(X_test)

# Evaluate models
svm_accuracy = accuracy_score(y_test, svm_pred)
dt_accuracy = accuracy_score(y_test, dt_pred)

print("SVM Accuracy:", svm_accuracy * 100)
print("Decision Tree Accuracy:", dt_accuracy * 100)
```

Listing 1: Core Implementation of SVM and Decision Tree

4.4 Model Configuration

The following parameters were used for the experiment.

Table 1: Model Configuration

Parameter	**Value**
Primary Model	SVM
Kernel	RBF
Regularization (C)	1.0
Gamma	scale
Baseline Model	Decision Tree
Train-Test Ratio	80:20
Random State	42
Feature Scaling	StandardScaler

5 Dataset and Experimental Results

5.1 Synthetic Dataset Description

The dataset contains 569 records distributed across six regions of Jharkhand:

- Ranchi
- Jamshedpur
- Dhanbad
- Dumka
- Hazaribagh

- Bokaro

The dataset contains 351 Benign records and 218 Malignant records.
The inspected CSV contains no missing values or duplicate rows.

Table 2: Features of the Synthetic Jharkhand Dataset

Feature	**Description**	**Type**
Patient_ID	Unique record identifier	Text
Region	Region of Jharkhand	Categorical
Age	Age in years	Numerical
Tumor_Size_mm	Tumor size in mm	Numerical
Lymph_Node_Status	Lymph node status	Binary
Family_History	Family history indicator	Binary
BMI	Body Mass Index	Numerical
Diagnosis	Benign/Malignant	Target

Patient_ID and Region are used for record identification and dataset analysis. They are not included as input features during model training.

5.2 Experimental Setup

The dataset was divided into training and testing subsets using an 80:20 ratio.

Table 3: Training and Testing Dataset Distribution

Dataset Subset	**Number of Records**
Total Records	569
Training Records	455
Testing Records	114

The SVM model was trained using the RBF kernel with $C = 1.0$ and $\gamma = \text{scale}$. The Decision Tree classifier was trained using the same training and testing records.

5.3 Observed Model Results

The final experimental results are presented below.

Table 4: Performance Comparison of Classification Models

Metric	**SVM (RBF)**	**Decision Tree**
Accuracy	**88.60%**	84.21%
Precision	**81.63%**	78.26%
Recall	**90.91%**	81.82%
F1-Score	**86.02%**	80.00%

The SVM classifier achieved an accuracy of 88.60%, while the Decision Tree baseline achieved 84.21%.

The SVM model obtained higher values across all four reported evaluation metrics on the selected test dataset.

5.4 Dataset Distribution Analysis

Figure 2 presents the distribution of the dataset across age, tumor size, diagnosis, and regions.

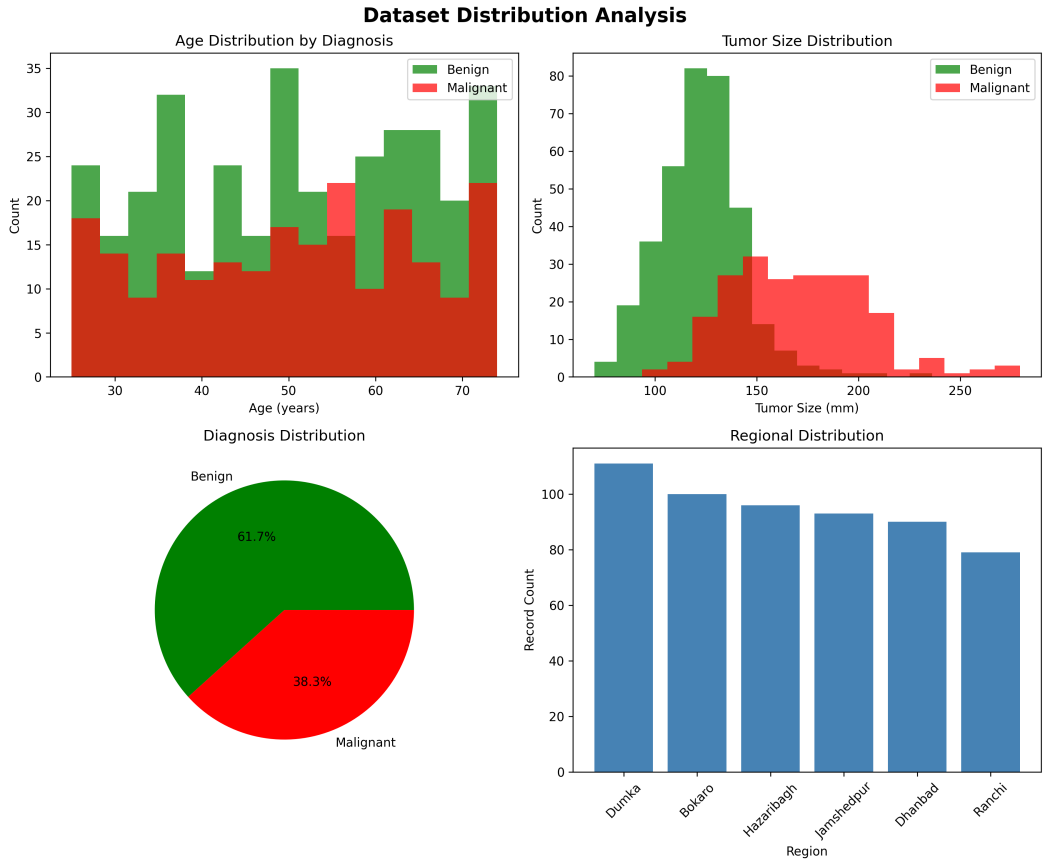


Figure 2: Dataset distribution showing age, tumor size, diagnosis, and regional distribution.

The figure provides an overview of the records used in the classification experiment. It includes the distribution of age and tumor size across the two diagnosis categories, the overall diagnosis distribution, and the number of records associated with each region.

5.5 Confusion Matrix Analysis

The confusion matrix provides a detailed view of the predictions made by the SVM classifier.

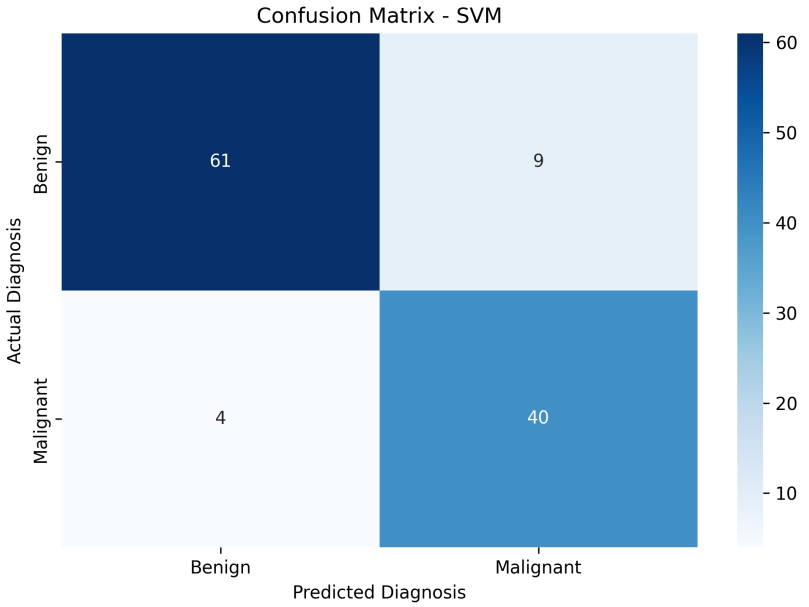


Figure 3: Confusion matrix of the trained SVM classifier.

The confusion matrix contains the following values:

Table 5: SVM Confusion Matrix Values

Classification Result	**Count**
True Negative (TN)	61
False Positive (FP)	9
False Negative (FN)	4
True Positive (TP)	40

The model correctly classified 61 Benign records and 40 Malignant records.

It incorrectly classified 9 Benign records as Malignant and 4 Malignant records as Benign.

The total number of correctly classified records is 101 out of 114 test records.

The resulting accuracy is:

$$Accuracy = \frac{61 + 40}{114} \times 100 \tag{13}$$

$$Accuracy = 88.60\% \tag{14}$$

5.6 Model Performance Comparison

The project compares two classification algorithms: SVM with an RBF kernel and Decision Tree.

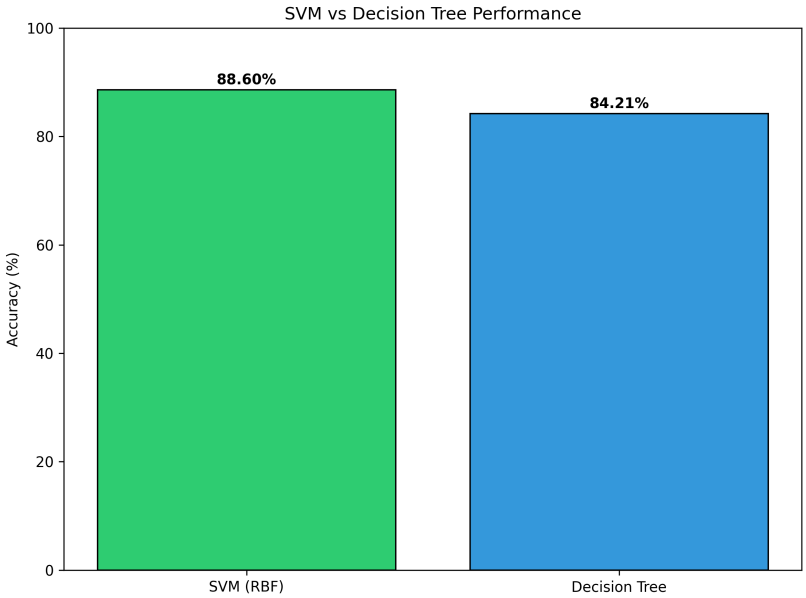


Figure 4: Accuracy comparison between SVM and Decision Tree classifiers.

The SVM model achieved 88.60% accuracy, compared with 84.21% for the Decision Tree baseline.

The results indicate that SVM performed better than the selected baseline on this dataset and experimental split.

5.7 Feature Correlation Analysis

Figure 5 presents the correlation heatmap of the selected input features and the encoded target variable.

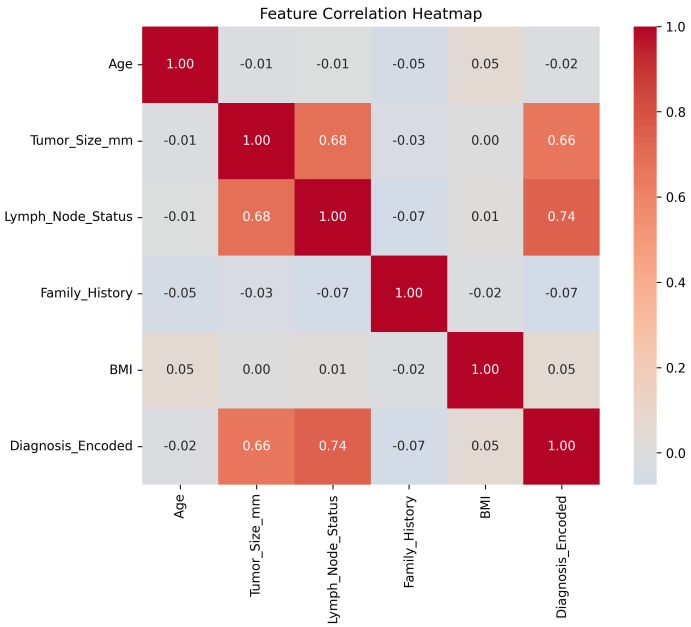


Figure 5: Correlation heatmap of the selected dataset features and encoded diagnosis.

The heatmap provides a numerical overview of the pairwise linear correlations between the selected variables.

It is used as a descriptive visualization of the synthetic dataset. Correlation values from this dataset should not be interpreted as evidence of clinical relationships.

5.8 Decision Boundary Visualization

To visualize the classification behavior of the RBF kernel, a separate two-dimensional SVM model was trained using Age and Tumor Size.

Figure 6 presents the resulting decision boundary.

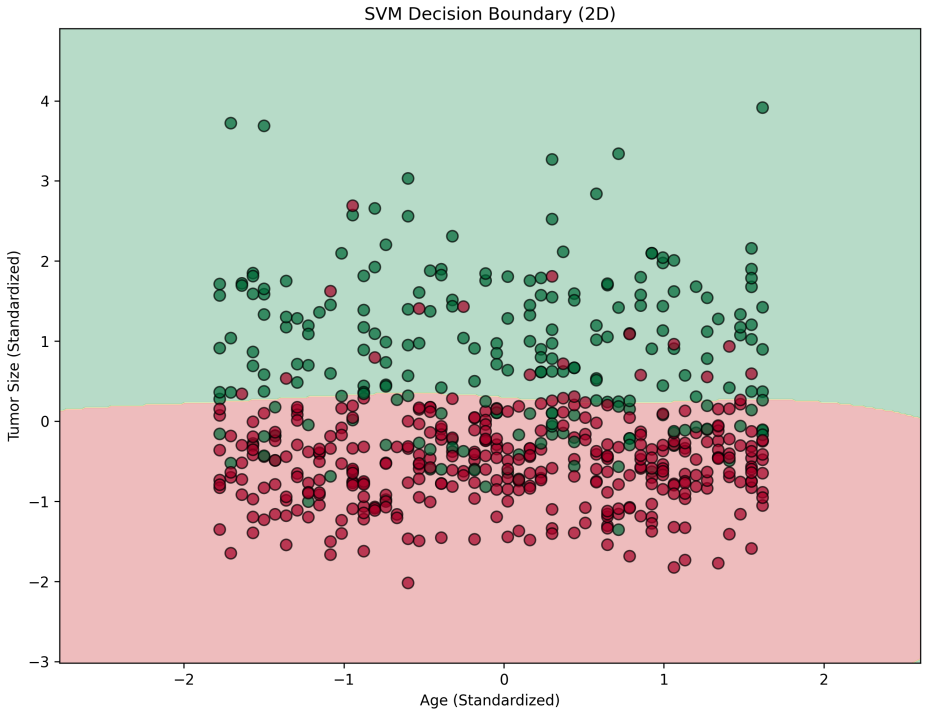


Figure 6: Two-dimensional SVM decision boundary using Age and Tumor Size features.

The visualization shows how the RBF kernel produces a non-linear decision boundary in the selected two-dimensional feature space.

It is important to note that this visualization uses only two features and is separate from the main SVM model, which uses all five input features.

5.9 Discussion of Results

The experimental results provide the following observations:

- **SVM Performance:** The SVM classifier achieved 88.60% accuracy on the test dataset.
- **Baseline Comparison:** The Decision Tree classifier achieved 84.21% accuracy.

- **Malignant Class Recall:** The SVM model achieved 90.91% recall for the Malignant class.
- **False Negatives:** The confusion matrix contains four false negative predictions.
- **Classification Comparison:** SVM achieved higher accuracy, precision, recall, and F1-score than the Decision Tree baseline in this experiment.

These findings describe the performance of the models on the selected synthetic dataset and 80:20 train-test split. They do not establish clinical effectiveness.

6 Complexity Analysis

6.1 Time and Space Complexity

Let:

- N be the number of training samples.
- D be the number of input features.
- N_s be the number of support vectors.

The computational complexity of SVM depends on the training algorithm, dataset size, and kernel.

Training Time:

The training time of kernel-based SVM can vary with the optimization implementation. A commonly used theoretical range is approximately:

$$O(N^2D) \text{ to } O(N^3D) \tag{15}$$

This is an approximate complexity description, not a measured execution-time result.

Prediction Time:

For a kernel SVM, prediction for one sample depends on the number of support vectors and the kernel computation cost. For the RBF kernel, it is approximately:

$$O(N_sD) \tag{16}$$

Space Complexity:

The input dataset requires approximately:

$$O(ND) \tag{17}$$

The support vectors require approximately:

$$O(N_sD) \tag{18}$$

6.2 Decision Tree Complexity

Decision Tree training commonly involves sorting and evaluating features at different nodes.

Its training complexity depends on the tree-building implementation, tree depth, and number of features.

For the present project, Decision Tree is used only as a baseline classifier. The comparison is based on predictive performance rather than measured execution time.

7 Conclusion and Limitations

7.1 Summary of Findings

This project implemented a Support Vector Machine classifier with an RBF kernel for classifying synthetic breast tumor records into Benign and Malignant categories.

The dataset contained 569 records representing six regions of Jharkhand. Five input features were selected for model training.

The SVM model achieved 88.60% accuracy, 81.63% precision, 90.91% recall, and 86.02% F1-score for the Malignant class.

The Decision Tree baseline achieved 84.21% accuracy, 78.26% precision, 81.82% recall, and 80.00% F1-score.

The results show that SVM performed better than the Decision Tree baseline on the selected synthetic dataset and test split.

7.2 Limitations

The primary limitation of this project is the use of synthetic data. The dataset does not contain real patient records collected from hospitals or clinical institutions.

Therefore, the experimental results cannot establish the clinical accuracy or real-world diagnostic effectiveness of the proposed model.

The dataset also contains a limited number of records and five selected input features. Further validation using independent real-world data would be necessary before considering any practical healthcare application.

7.3 Future Scope

The project can be extended in the following ways:

- Evaluate the model using real and anonymized patient data collected with appropriate ethical approvals.
- Increase the dataset size and include additional clinically relevant features.
- Apply cross-validation and systematic hyperparameter optimization.
- Develop a web-based interface for demonstrating the classification workflow.
- Perform independent validation before considering any clinical application.

7.4 Concluding Remarks

This project demonstrates the implementation of an SVM-based classification system using a synthetic regional dataset.

The RBF kernel allows the SVM model to represent non-linear decision boundaries, while feature scaling provides standardized input values for training.

The experimental comparison shows that SVM achieved better classification metrics than the Decision Tree baseline on the selected test split.

The results are limited to the synthetic dataset used in this project and should not be interpreted as a substitute for clinical diagnosis.

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